

Graphs, Differentiation & Functions

Practical graphs, curve sketching, calculus and function notation.



Use these notes well and you will be able to:

- Interpret and draw travel (distance–time) and speed–time graphs.
- Construct tables of values and draw graphs of linear, quadratic, cubic, reciprocal, exponential.
- Sketch curves: identify roots, turning points, asymptotes, and symmetry.
- Use differentiation to find gradients and stationary points (F 02).
- Use function notation; find inverse and composite functions (F 02).

If you follow the steps, you will succeed.

Practical Graphs

GRAPH TYPE	GRADIENT MEANS	AREA UNDER GRAPH MEANS
Distance–time	Speed (gradient = speed)	—
Speed–time	Acceleration	Distance travelled
Velocity–time	Acceleration (signed)	Displacement
Conversion graph	Exchange / conversion rate	—
Any curve (E)	Gradient of tangent at pt	Area using trapezium rule

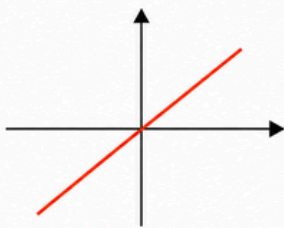
Tip: Distance–time: flat line = stationary. Steep line = fast. Speed–time: area of triangle = $\frac{1}{2}bh$ = distance. Area of trapezium = $\frac{1}{2}(a+b)h$ = distance. Negative gradient on speed–time = deceleration.

Key Words

KEY WORD	SIMPLE DEFINITION
Gradient	Steepness of a line/curve at a point: $m = \text{change in } y \div \text{change in } x$.
y–intercept	Where the graph crosses the y–axis ($x = 0$).
Root / zero	Where the graph crosses the x–axis ($y = 0$). Solving $f(x) = 0$.
Turning point	Where gradient = 0 — a maximum (peak) or minimum (valley).
Asymptote	A line that a curve approaches but never reaches.
Exponential	Of the form $y = ab^x$ — grows/decays rapidly.
Tangent to curve	A straight line touching the curve at exactly one point — gradient = dy/dx .
Stationary point	Where $dy/dx = 0$ — the gradient is zero at that instant.
Function $f(x)$	A rule mapping each x–value to exactly one y–value.
Domain	The set of allowed input (x) values for a function.
Range	The set of possible output (y) values for a function.
Inverse $f^{-1}(x)$	Reverses the function: if $f(a)=b$ then $f^{-1}(b)=a$.
Composite $fg(x)$	f applied after g: $fg(x) = f(g(x))$. Apply g first, then f.

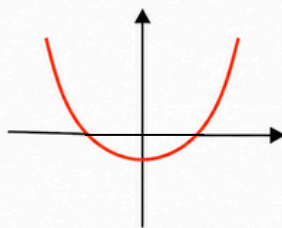
Graphs of Functions

For each function type: construct a table of values, plot points, join with a smooth curve.



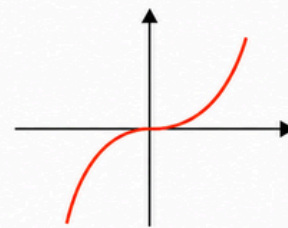
Straight line. m = gradient. c = y -intercept.

Linear $y = ax + b$



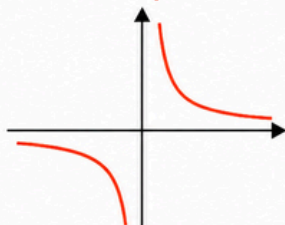
Parabola. Opens up ($a > 0$) or down ($a < 0$).

Quadratic $y = ax^2 + bx + c$



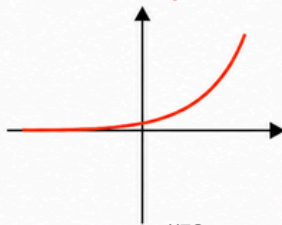
S-shaped curve. Up to 3 roots, 2 turning points.

Cubic $y = ax^3 + bx^2 + cx$



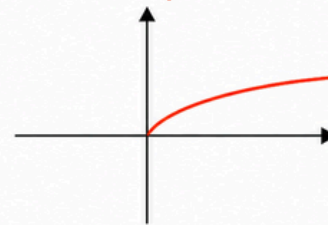
Two branches. Asymptotes: $x=0$ and $y=0$.

Reciprocal $y = a/x$



Rapid growth/decay. Horizontal asymptote at $y=c$.

Exponential $y = ab^x + c$



Square root curve. Only for $x \geq 0$. F 02.

$y = ax^{1/2} + b$

Sketching Curves — Key Features

CURVE	ROOTS ($y=0$)	TURNING PTS	ASYMPTOTES / SYMMETRY
$y = ax + b$	$x = -b/a$	None	Line of symmetry: none
$y = ax^2 + bx + c$	0, 1, or 2	1 vertex	Axis of sym: $x = -b/2a$
$y = ax^3 + \dots$	Up to 3	Up to 2	Point symmetry if only $x^3 + c$
$y = a/x$	None	None	$x=0$ and $y=0$ (both axes)
$y = ab^x + c$	May have one	None	Horizontal: $y=c$

Differentiation [F 02]

Differentiation finds the gradient function dy/dx for any point on a curve. Stationary points occur where $dy/dx = 0$. A stationary point is a maximum if $d^2y/dx^2 < 0$, a minimum if $d^2y/dx^2 > 0$.

DIFFERENTIATION RULE (for ax^n) Sum of terms: differentiate each term separately. Constant term disappears.

$dy/dx (ax^n) = anx^{(n-1)}$. You multiply by power (n), reduce power by 1

FUNCTION $f(x)$	DERIVATIVE $f'(x)$ ($= dy/dx$)	WHAT IT GIVES
$5x^3$	$15x^2$	Gradient at any x
$3x^2 - 4x + 7$	$6x - 4$	Gradient function
$x + 2x^3 - x$	$4x^3 + 6x^2 - 1$	Gradient function
7	0	Constants disappear

Find the gradient of $y = x^3 - 6x^2 + 9x + 2$ at $x = 1$

$$dy/dx = 3x^2 - 12x + 9$$

$$\text{At } x=1: 3(1) - 12(1) + 9 = 3 - 12 + 9 = 0$$

Gradient = 0 ($x=1$ is a stationary point)

Find and classify the stationary points of $y = 2x^3 - 9x^2 + 12x$

$$dy/dx = 6x^2 - 18x + 12 = 6(x^2 - 3x + 2) = 6(x-1)(x-2)$$

$$\text{Set } = 0: x = 1 \text{ or } x = 2$$

$$d^2y/dx^2 = 12x - 18$$

$$\text{At } x=1: 12-18 = -6 < 0 \text{ MAXIMUM. } y = 2-9+12 = 5$$

$$\text{At } x=2: 24-18 = +6 > 0 \text{ MINIMUM. } y = 16-36+24 = 4$$

Max at (1, 5); Min at (2, 4)

Functions — Notation, Inverse, Composite [F 02]

Given $f(x) = 3x - 5$. Find $f(4)$ and $f^{-1}(x)$.

$$f(4) = 3(4) - 5 = 7$$

$$\text{Inverse: let } y = 3x - 5 \quad y + 5 = 3x \quad x = (y + 5)/3$$

$$\text{Replace } y \text{ with } x: f^{-1}(x) = (x + 5)/3$$

$$\mathbf{f(4) = 7 \quad f^{-1}(x) = (x + 5) / 3}$$

$f(x) = 3/(x+2)$, $g(x) = (3x+5)^2$. Find $fg(x)$ as a single fraction.

$$fg(x) = f(g(x)) \text{ — apply } g \text{ first, then } f$$

$$g(x) = (3x+5)^2$$

$$fg(x) = 3/((3x+5)^2+2) = 3/(9x^2+30x+25+2)$$

$$\mathbf{fg(x) = 3 / (9x^2 + 30x + 27) = 1 / (3x^2 + 10x + 9)}$$

Practice Questions

PRACTICAL GRAPHS AND SKETCHING

1. A car travels at 30m/s for 5s, then decelerates to 0 m/s over 4 s. Sketch the speed–time graph and find the total distance.
2. Sketch $y = x^2 - 4x - 5$. Mark the roots and vertex.
3. Sketch $y = 2/x$. State the equations of the asymptotes.

DIFFERENTIATION [F 02]

$$4. \text{ Differentiate: } y = 4x^3 - 5x^2 + 3x - 7$$

$$5. \text{ Find the gradient of } y = x^3 - 4x + 1 \text{ at } x = 2.$$



Composite functions: right to left. $fg(x)$ means do g FIRST, then f .

Keep going — see you in the next lesson! From FREDi